

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

Finding reasoning paths :

- 1 . The anatomical snuff box is an area where several important arteries and nerves are located .
- 2 . The brachial artery runs along the inner border of the anatomical snuff box and gives rise to the radial artery .
- 3 . Damage to the brachial artery in this area could lead to excessive bleeding due to its anatomical position and function .
- 4 . The anatomical snuff box is also associated with the median nerve , but the primary concern in this scenario is arterial damage , not nerve injury .

Reasoning Process :

- 1 . The anatomical snuff box is a depression on the lateral aspect of the forearm , located between the proximal bicipital groove and the base of the styloid process of the ulna .
- 2 . This area is significant because it houses several important structures , including the brachial artery , which is the main artery supplying blood to the upper limb .
- 3 . The brachial artery gives rise to the radial artery , which is responsible for the pulse in the wrist and provides blood to the forearm and hand .
- 4 . In a penetrating stab wound , the brachial artery is particularly vulnerable in this location due to its position and the limited space for collateral circulation .
- 5 . Damage to the brachial artery can lead to uncontrollable bleeding , especially if the wound is deep enough to reach the arterial structures .
- 6 . The presence of the median nerve in this area does not directly contribute to the bleeding , as the nerve is not primarily responsible for the arterial flow in this region .

Conclusion :

Damage to the brachial artery is most likely responsible for the excessive bleeding from the right hand . This conclusion is based on the anatomical location of the artery in the anatomical snuff box and its role in providing blood to the forearm and hand .

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<answer>

Brachial artery

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